HW5 Report: Hierarchical Clustering, DBSCAN, and Cluster Evaluation

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Clustering methods behave differently because they impose different ideas of what a cluster should look like. This report follows that idea as a running thread. We begin with an easy setting where most methods should agree, then move toward datasets where bridging points, density variation, and curved geometry make those assumptions visible in the output.

Task 1: Hierarchical Clustering on Well-Separated Clusters

The first dataset gives a clean starting point: two compact clouds with a visible gap between them. In this setting, the main question is whether the linkage rule matters at all once the groups are already well separated. Figure 1 shows that single, complete, and average linkage all produce the same final partition at $K = 2$. Each method gives a balanced 200–200 split, and the pairwise Rand agreement is 1.000 throughout. The dendrogram heights differ, but the final grouping does not. This is a useful baseline because it shows hierarchical clustering at its most stable.

Single linkage uses the minimum cross-cluster distance, complete linkage uses the maximum, and average linkage uses the mean cross-cluster dissimilarity. When there is a wide empty gap between two compact groups, all three notions induce the same top-level split even though the intermediate merge order changes. Figure 2 makes that point visible: the final merge happens far above the within-cluster merges, so cutting the dendrogram anywhere in that gap still returns the same two-cluster solution.

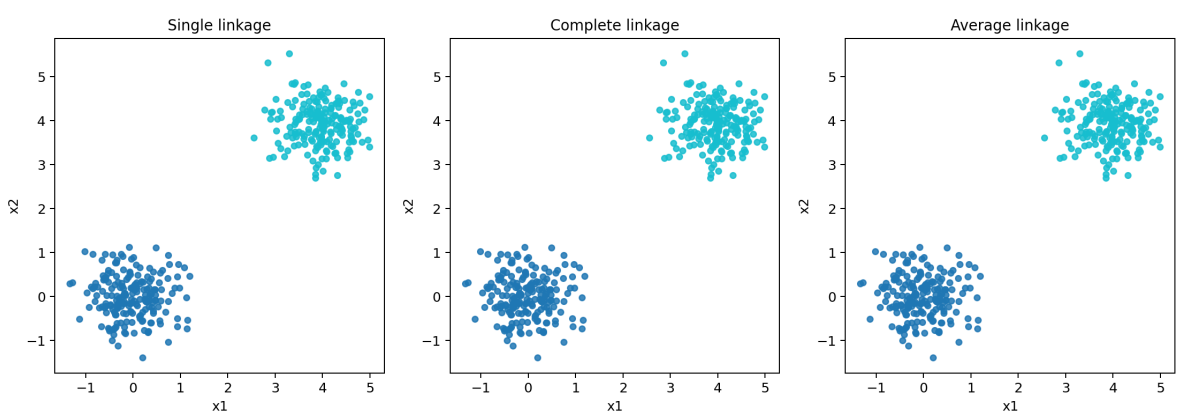

Figure 1: Cluster assignments under single, complete, and average linkage. Although the merge paths differ, all three methods recover the same two-cluster split because the two groups are clearly separated.

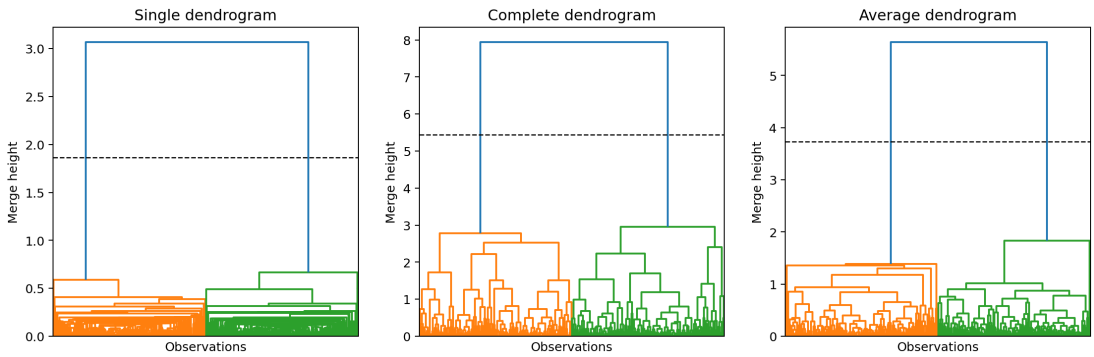

Figure 2: Dendrograms for single, complete, and average linkage. The last merge occurs at clearly larger heights than the within-cluster merges, which explains why all three linkage rules agree after cutting at $K = 2$.

Task 2: The Chaining Effect in Hierarchical Clustering

The next dataset adds a thin bridge between two dense blobs. This small change is enough to separate the linkage rules. Figure 3 shows that single linkage almost collapses the dataset into one cluster, with sizes cluster 1=349, cluster 2=1. Complete linkage remains balanced at cluster 1=174, cluster 2=176. DBSCAN produces noise=14, cluster 1=179, cluster 2=157. The reason is visible in the figure: single linkage only needs a short sequence of nearby points, so the bridge acts like a connector. Complete linkage is stricter because it also reflects the farthest-point distance inside a merged cluster, so it resists this elongated connection.

A cluster can grow through a sequence of locally close points even if the resulting object has large overall diameter. The bridge points are not dense enough to form a stable compact cluster on their own, but they are close enough pairwise for single linkage to keep merging across the gap. DBSCAN sits between these two outcomes: it respects local density connectivity and therefore captures the two dense ends, while still identifying some low-support bridge points as noise instead of forcing a full merge.

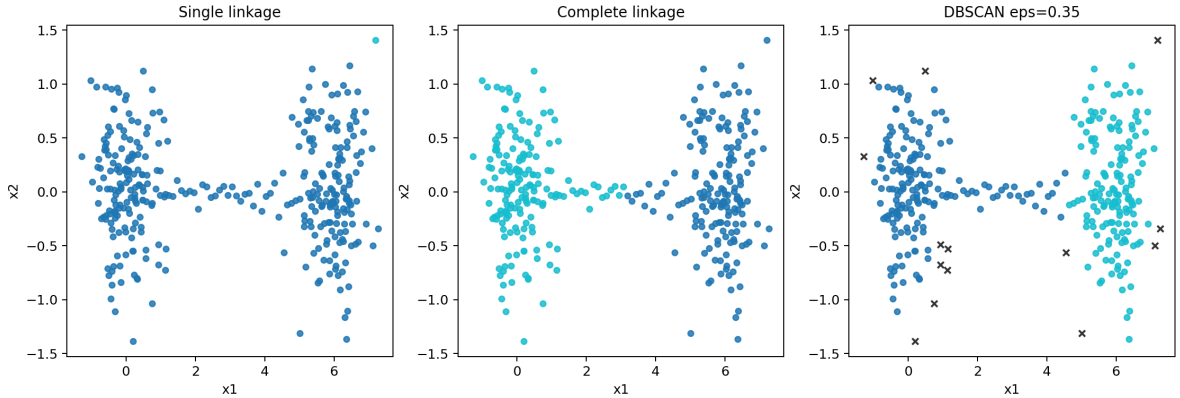

Figure 3: Comparison of clustering under a bridging structure. Single linkage follows the chain and nearly merges everything, while complete linkage preserves two compact groups and DBSCAN keeps most of the bridge as belonging to one of the cluster as they are adjacent and dense. It also tags some points as noise.

Task 3: Density-Based Clustering with DBSCAN

After linkage behavior, the focus shifts to density. The two clusters in this task are not equally dense, so a single global choice of ε is likely to favor one side more than the other. Figure 4 confirms that pattern. At $\varepsilon = 0.2$, the tight left blob is recovered cleanly but the diffuse right blob fragments, leaving 188 points as noise. At $\varepsilon = 0.5$, the left blob still looks stable, while the right side becomes one main cluster of 267 points plus a small fragment and 21 noise points. At $\varepsilon = 1.0$, the method finally produces two main clusters of size 300 and 298 with only 2 noise points. The progression shows that the failure is not random. It comes from using one density scale for two regions that spread differently.

The core-border-noise distinction is useful here. For small ε, many points in the diffuse cluster fail to accumulate enough neighbors inside their local neighborhoods, so they are labeled noise even though they visibly belong to the same broad cloud. Increasing ε converts more of those points into border or core points, but the same increase would be unnecessary for the dense left cluster. That mismatch explains the standard limitation of DBSCAN in variable-density data: one global density threshold cannot simultaneously fit both a tight cluster and a much more dispersed one.

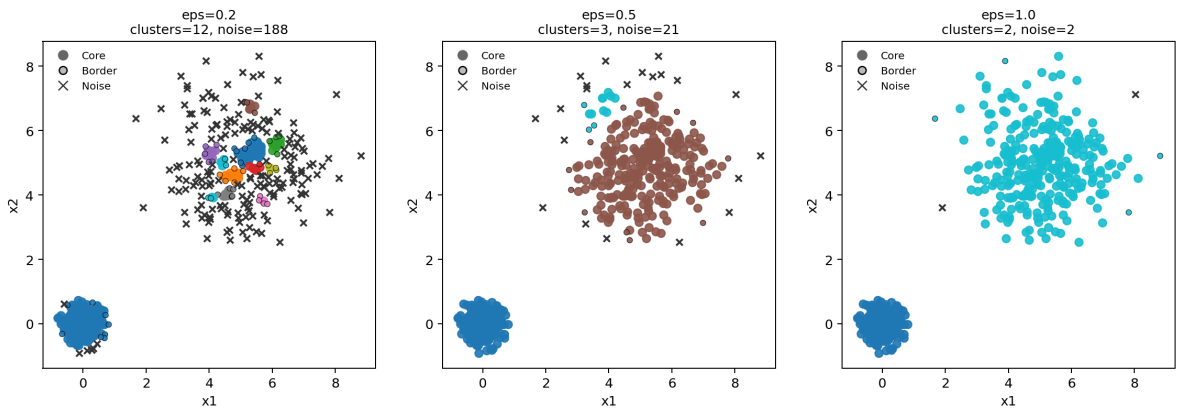

Figure 4: DBSCAN results across the required ε values. Core points form the dense backbone, border points sit near the edge of a cluster, and noise points reveal where the chosen density threshold is too strict.

Task 4: Intrinsic Clustering Evaluation

Once the clustering structure is visually simple again, it becomes possible to compare evaluation metrics more directly. The goal here is to see whether internal and external measures tell a consistent story across different values of K. Figure 5 and Table 1 show a strong agreement: every metric selects $K = 3$. The silhouette score is highest at $K = 3$, the Davies–Bouldin index is lowest at $K = 3$, and the Calinski–Harabasz score peaks there as well. The extrinsic measures also align with that choice, with Rand $= 0.9933$ and Fowlkes–Mallows $= 0.9900$. In this dataset, under-clustering at $K = 2$ and over-splitting at $K = 4, 5$ are both clearly penalized.

A useful extra observation is that the three intrinsic criteria emphasize different ideas of a good partition. Silhouette rewards points being closer to their own cluster than to neighboring clusters, Davies–Bouldin penalizes large within-cluster spread relative to separation, and Calinski–Harabasz compares between-cluster variance to within-cluster variance. Their agreement at $K = 3$ is therefore stronger than one score alone because it says the same partition is simultaneously compact, well

separated, and variance efficient.

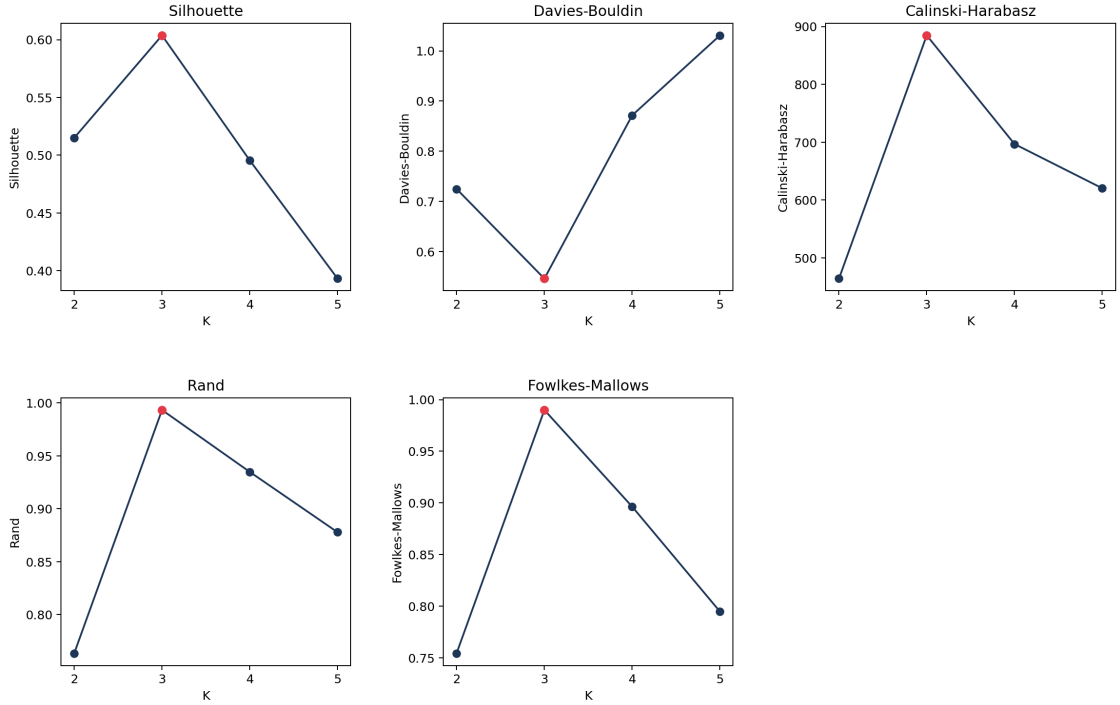

Figure 5: Task 4 metric trends across $K = 2, 3, 4, 5$. All the curves identify $K = 3$ as the most suitable partition.

K	Silhouette	Davies-Bouldin	Calinski-Harabasz	Rand	Fowlkes-Mallows
2	0.5151	0.7250	464.09	0.7635	0.7546
3	0.6038	0.5464	884.79	0.9933	0.9900
4	0.4955	0.8716	696.84	0.9348	0.8964
5	0.3932	1.0309	620.36	0.8778	0.7949

Table 1: Task 4 evaluation scores. The intrinsic and extrinsic criteria all point to the same value, making this a good example where metric-based model selection is stable.

Task 5: Clustering Nonlinearly Separable Data (Concentric Circles)

The report then moves to a dataset where visual compactness and true cluster structure are no longer the same thing. In the concentric circles data, the correct grouping is based on rings, not on convex partitions. Figure 6 and Table 2 show that KMeans and GMM both fail in a similar way: each creates a radial split and gets a Rand score near 0.50. Their silhouette scores are around 0.35, which is much higher than the correct ring-based solution. In contrast, agglomerative clustering with single linkage and DBSCAN recover the two circles exactly, giving Rand and Fowlkes–Mallows values of 1.0000. The low silhouette near 0.113 for those correct solutions is informative rather than problematic. It shows that Euclidean compactness is not a reliable guide for nested circular structure.

This task also sharply separates the assumptions behind the algorithms. KMeans prefers Voronoi-style partitions around centroids, and GMM prefers ellipsoidal components, so both methods cut across the rings. Single-linkage agglomeration and DBSCAN instead rely on connectivity: nearby points along the same ring can be chained together even when the cluster is not convex. The figure is therefore a good reminder that an apparently worse intrinsic score can correspond to a much better recovery of the true geometry.

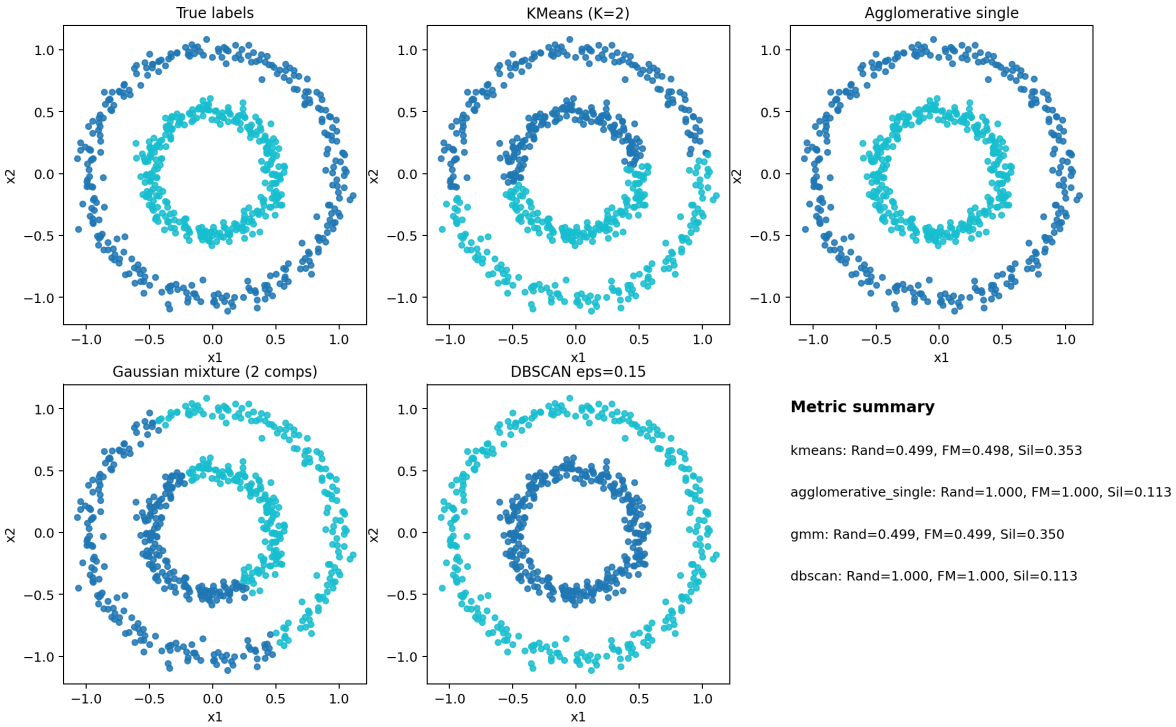

Figure 6: Task 5 clustering results on concentric circles. Methods that assume compact or model-based elliptical structure split the rings radially, while connectivity-based methods recover the circular geometry.

Method	Rand	Fowlkes-Mallows	Silhouette	Clusters	Noise
kmeans	0.4992	0.4983	0.3531	2	0
agglomerative single	1.0000	1.0000	0.1131	2	0
gmm	0.4992	0.4986	0.3504	2	0
dbscan	1.0000	1.0000	0.1131	2	0

Table 2: Task 5 evaluation scores across methods. The table highlights the contrast between internal compactness-based preference and external agreement with the true ring labels.

Task 6: Clustering Nonlinearly Separable Data (Spiral Data)

The final dataset pushes the geometry further. The spiral arms are curved, non-convex, and not equally spaced from the center outward. The task file explicitly asks for DBSCAN and then refers to both methods, so I use KMeans as the distance-based comparison. Figure 8 and Table 3 show that neither method fully solves the problem under the required setup. KMeans cuts across the arms and gives Rand = 0.5510. The best DBSCAN setting within the required grid is $\varepsilon = 0.5$ and minPts = 3, with Rand = 0.6205, but that run still produces 73 clusters and 280 noise points. So DBSCAN captures local curved pieces better than KMeans, yet it still fragments the spirals because one fixed density threshold cannot adapt smoothly from the dense inner turns to the sparse outer turns.

Figure 7 helps explain why tuning is difficult. Small ε values break the arms into many short fragments and label most outer points as noise, while large ε values begin to connect points across neighboring turns that should remain separate. Increasing minPts makes the density requirement even harder to satisfy in the sparse outer region. Here we see that the geometry and density profile of the spiral violate the assumptions of both centroid-based partitioning and single-scale density thresholding.

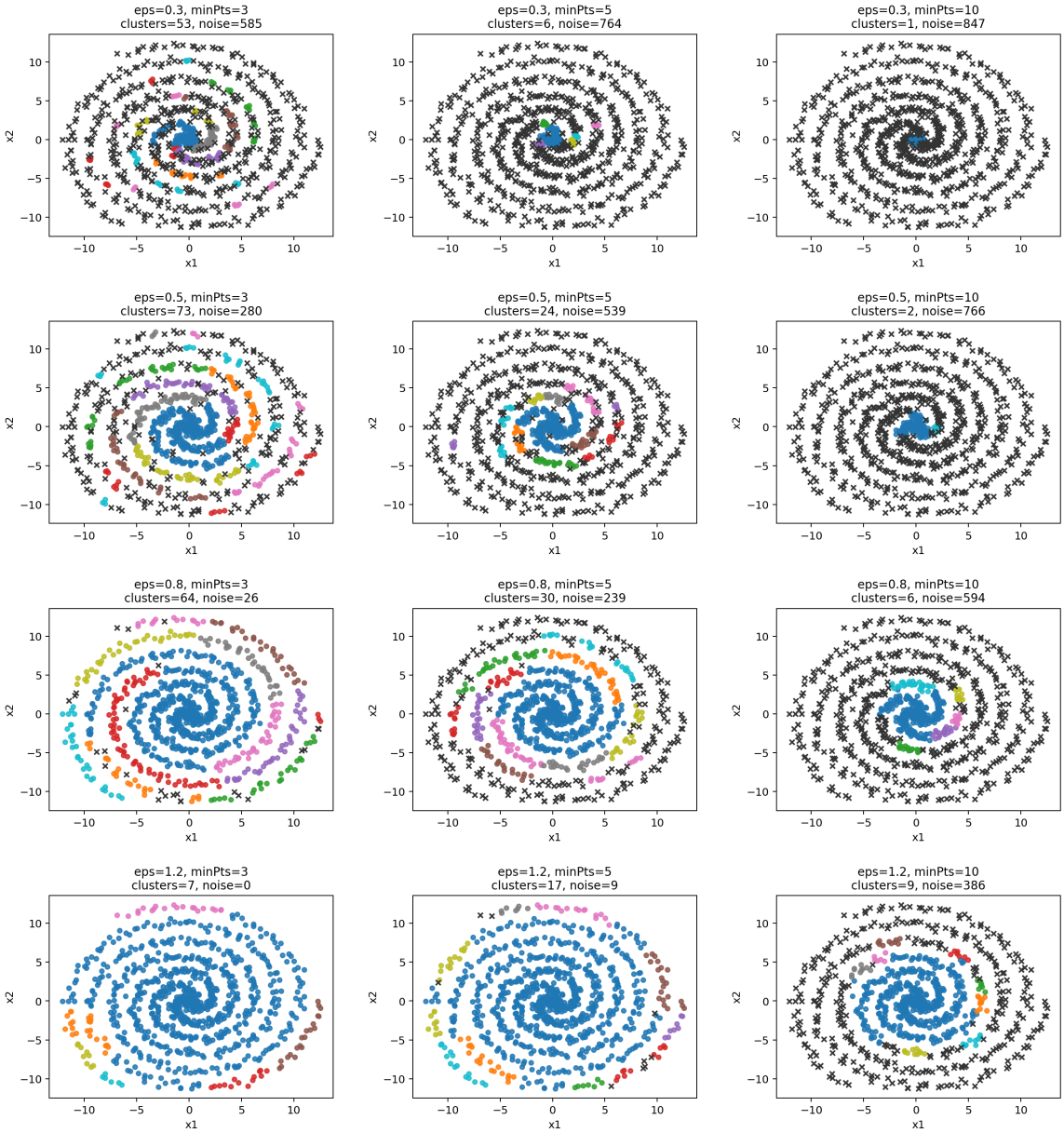

Figure 7: DBSCAN parameter grid for the spiral data. The grid shows the trade-off between fragmentation at small ε, heavy noise at large minPts, and undesired cross-arm connections as neighborhoods grow.

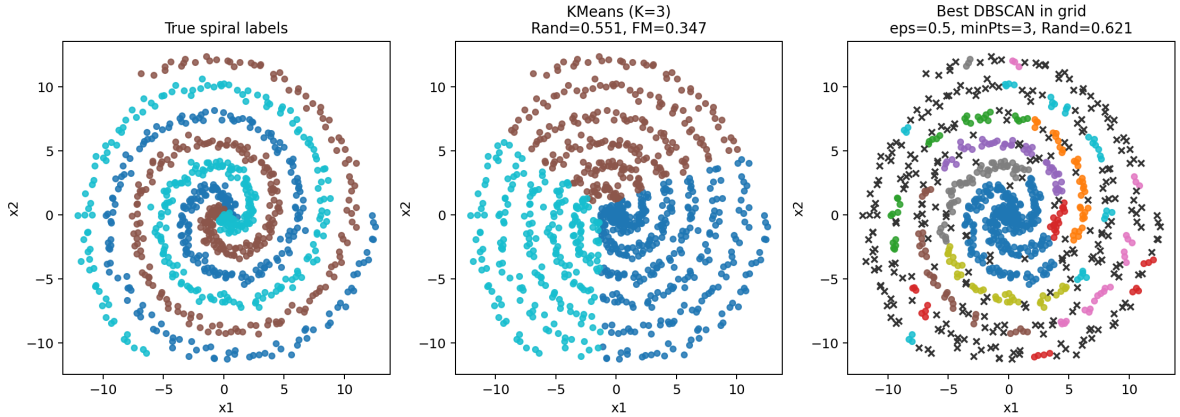

Figure 8: Task 6 comparison on the spiral dataset. KMeans forms wedge-like regions that cut across the arms, while the best DBSCAN setting follows some local curved pieces but breaks the full spirals into many fragments.

ε	minPts	Clusters	Noise	Rand	Fowlkes-Mallows
0.3	3	53	585	0.5226	0.3834
0.3	5	6	764	0.4232	0.4938
0.3	10	1	847	0.3696	0.5436
0.5	3	73	280	0.6205	0.2422
0.5	5	24	539	0.5338	0.3721
0.5	10	2	766	0.4184	0.4971
0.8	3	64	26	0.6007	0.3083
0.8	5	30	239	0.5851	0.3063
0.8	10	6	594	0.5017	0.4106
1.2	3	7	0	0.3999	0.5212
1.2	5	17	9	0.4326	0.4881
1.2	10	9	386	0.5244	0.3782

Table 3: Task 6 DBSCAN results across the required parameter grid. The table shows that higher agreement within the grid still comes with heavy fragmentation, which is visible in the comparison figure.

Conclusion

Taken together, the six tasks form a gradual shift from easy separation to structurally difficult data. The early tasks show when standard linkage rules and metric selection behave predictably. The later tasks show where those same tools become sensitive to chaining, density mismatch, or non-convex geometry. The main lesson is that clustering performance depends less on one method being globally best and more on whether the method assumptions match the actual structure in the data.